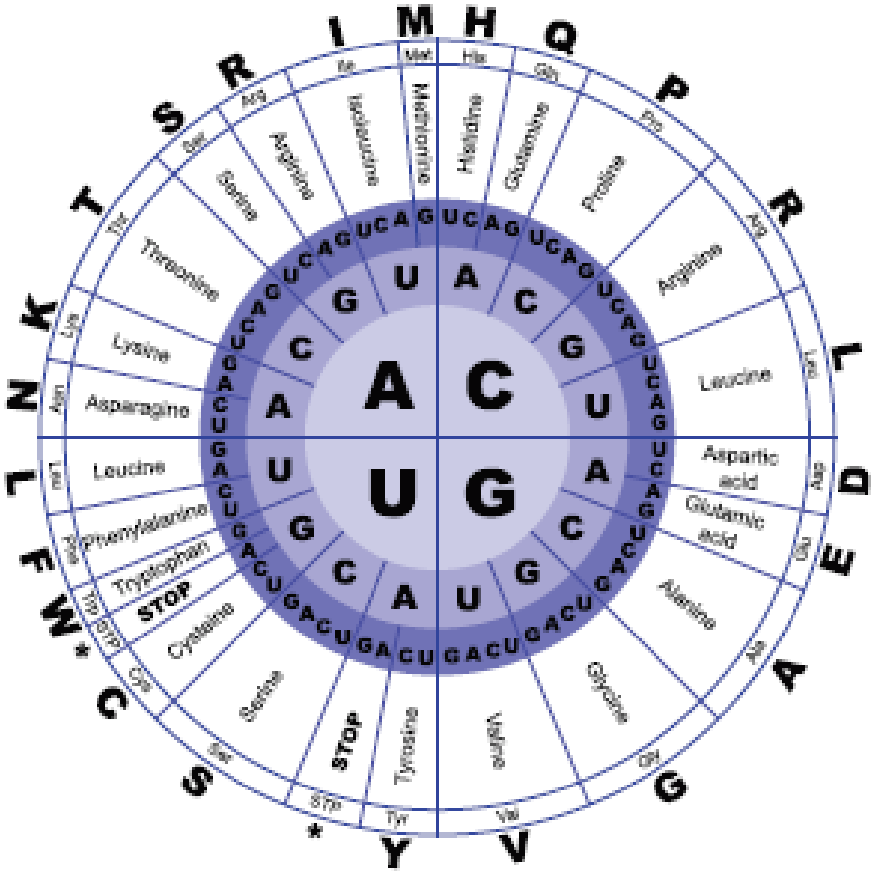
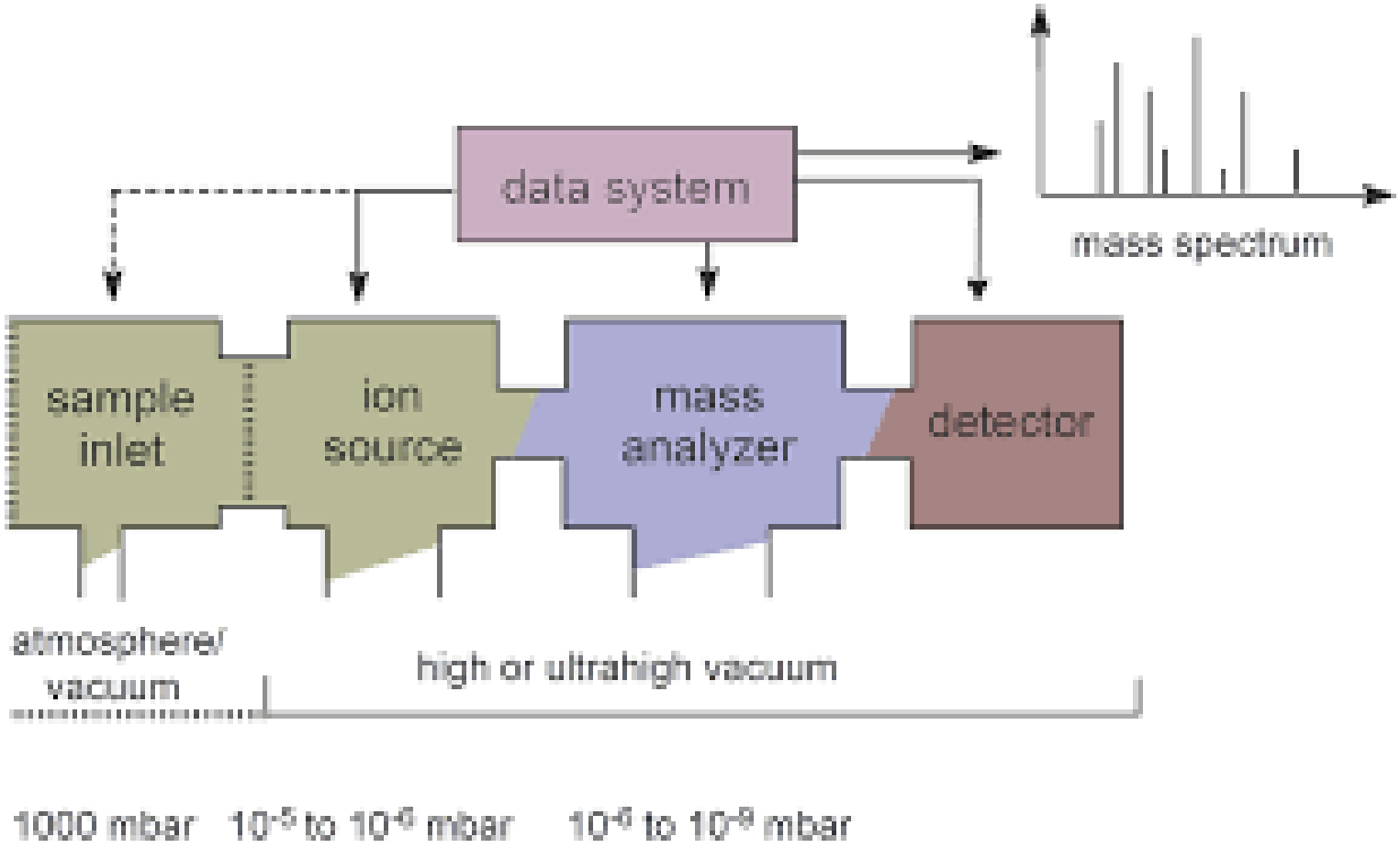
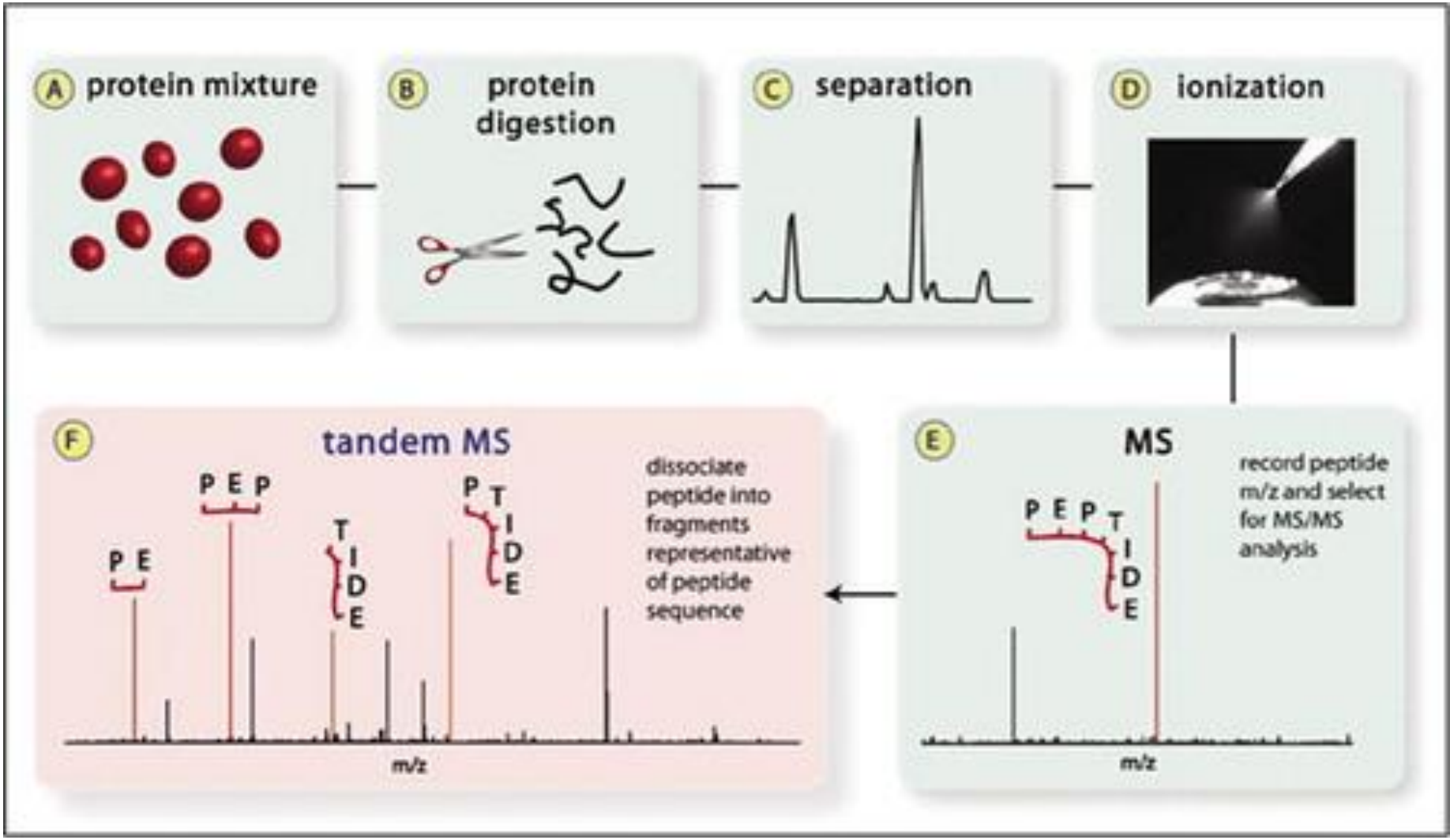
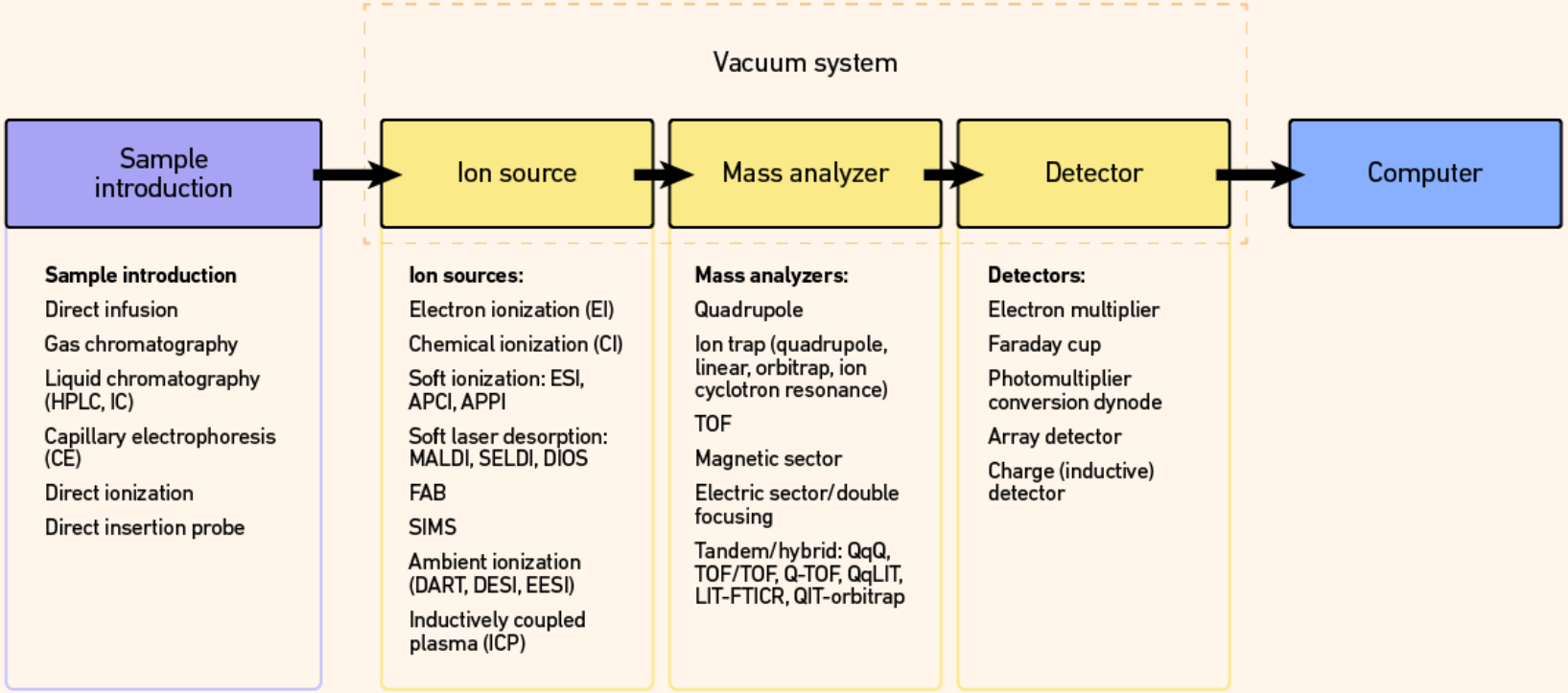
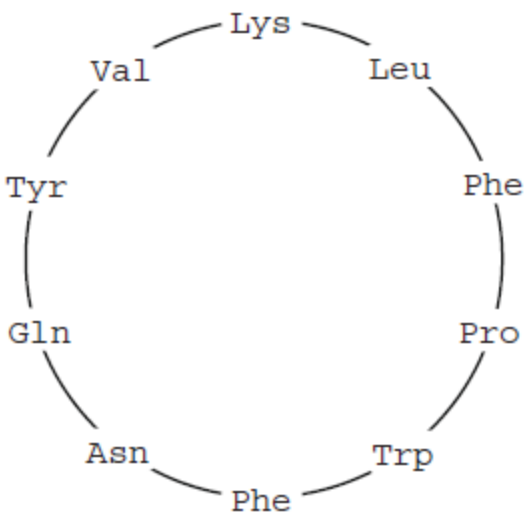


FIGURE 4.1 The genetic code describes the translation of an RNA 3-mer (codon) into one of 20 amino acids. The first three circles, moving from the inside out, represent the first, second, and third nucleotides of a codon. The fourth, fifth, and sixth circles define the translated amino acid in three ways: the amino acid's full name, its 3-letter abbreviation, and its single-letter abbreviation. Three of the 64 total RNA codons are stop codons, which halt translation.


data system
mass spectrum
sample inlet
ion source
mass analyzer
detector
atmosphere/ vacuum
high or ultrahigh vacuum
1000 mbar
10^{-5} to 10^{-6} mbar
10^{-6} to 10^{-8} mbar


A protein mixture
B protein digestion
C separation
D ionization
E MS
record peptide m/z and select for MS/MS analysis
P E P T I D E
m/z
F tandem MS
dissociate peptide into fragments representative of peptide sequence
P E
P E P
T I D E
P T I D E
m/z


Vacuum system
Sample introduction
Ion source
Mass analyzer
Detector
Computer
Sample introduction
Direct infusion
Gas chromatography
Liquid chromatography (HPLC, IC)
Capillary electrophoresis (CE)
Direct ionization
Direct insertion probe
Ion sources:
Electron ionization (EI)
Chemical ionization (CI)
Soft ionization: ESI, APCI, APPI
Soft laser desorption: MALDI, SELDI, DIOS
FAB
SIMS
Ambient ionization (DART, DESI, EESI)
Inductively coupled plasma (ICP)
Mass analyzers:
Quadrupole
Ion trap (quadrupole, linear, orbitrap, ion cyclotron resonance)
TOF
Magnetic sector
Electric sector/double focusing
Tandem/hybrid: QqQ, TOF/TOF, Q-TOF, QqLIT, LIT-FTICR, QIT-orbitrap
Detectors:
Electron multiplier
Faraday cup
Photomultiplier conversion dynode
Array detector
Charge (inductive) detector



1 Val-Lys-Leu-Phe-Pro-Trp-Phe-Asn-Gln-Tyr
2 Lys-Leu-Phe-Pro-Trp-Phe-Asn-Gln-Tyr-Val
3 Leu-Phe-Pro-Trp-Phe-Asn-Gln-Tyr-Val-Lys
4 Phe-Pro-Trp-Phe-Asn-Gln-Tyr-Val-Lys-Leu
5 Pro-Trp-Phe-Asn-Gln-Tyr-Val-Lys-Leu-Phe
6 Trp-Phe-Asn-Gln-Tyr-Val-Lys-Leu-Phe-Pro
7 Phe-Asn-Gln-Tyr-Val-Lys-Leu-Phe-Pro-Trp
8 Asn-Gln-Tyr-Val-Lys-Leu-Phe-Pro-Trp-Phe
9 Gln-Tyr-Val-Lys-Leu-Phe-Pro-Trp-Phe-Asn
10 Tyr-Val-Lys-Leu-Phe-Pro-Trp-Phe-Asn-Gln

FIGURE 4.4 Tyrocidine B1 is a cyclic peptide (left), and so it has ten different linear representations (right).

G	A	S	P	V	T	C	I	L	N	D	K	Q	E	M	H	F	R	Y	W
57	71	87	97	99	101	103	113	113	114	115	128	128	129	131	137	147	156	163	186

FIGURE 4.6 Integer masses of amino acids. The integer mass of each amino acid is computed by adding the number of protons and neutrons in the amino acid molecule.

Brute Force Cyclopeptide Sequencing

The mass of the entire peptide is usually known.

Algorithm:

1. Generate all **peptides** with given mass (1322).
2. Form their theoretical spectra.
3. Look for matches with the given **spectrum**

Brute Force Algorithm: "Try all" candidate solutions of a given kind.

Why Is Brute Force So Bad?

114–128–129–113	101–131–115–137
N Q E L	T M D H
Total Mass: **484**	Total Mass: **484**

These peptides are *completely different*.

Why Is Brute Force So Bad?

Spectrum of **TMDH**

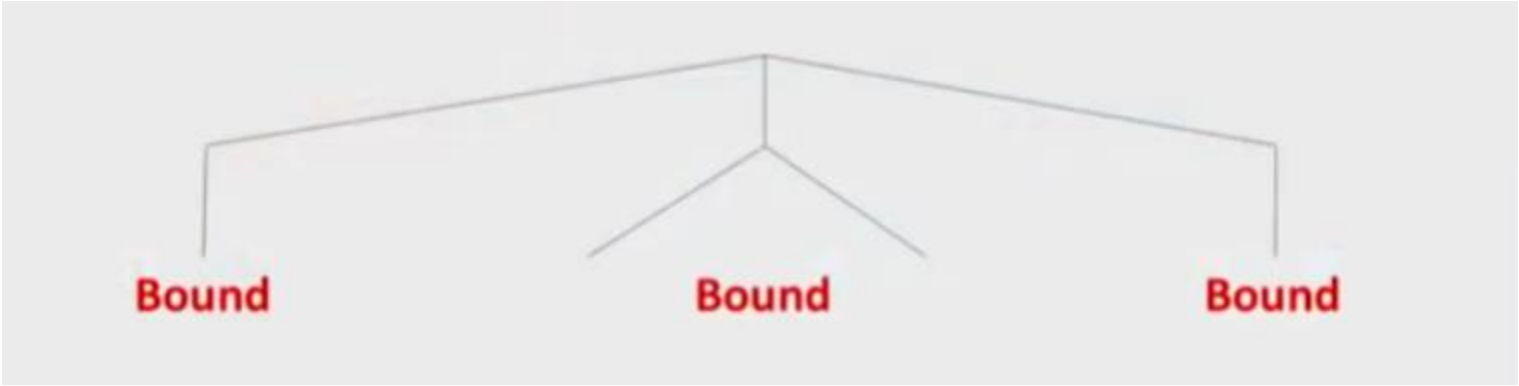
""	0
T	101
D	115
M	131
H	137
TM	232
HT	238
MD	246
DH	252
TMD	347
DHT	353
HTM	369
MDH	383
TMDH	484

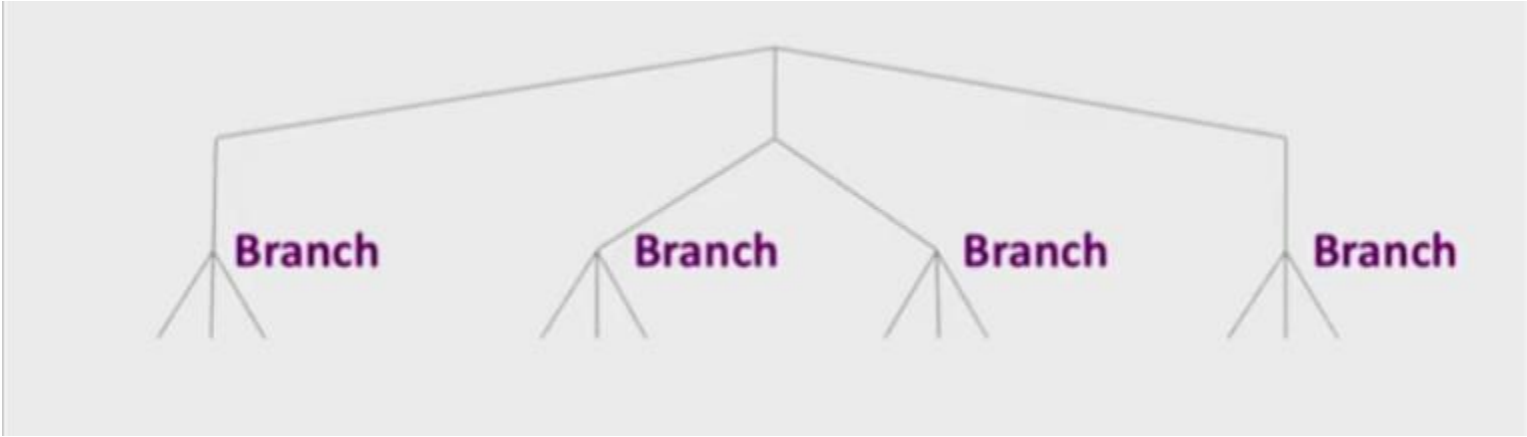
Their spectra completely disagree!

Spectrum of **NQEL**

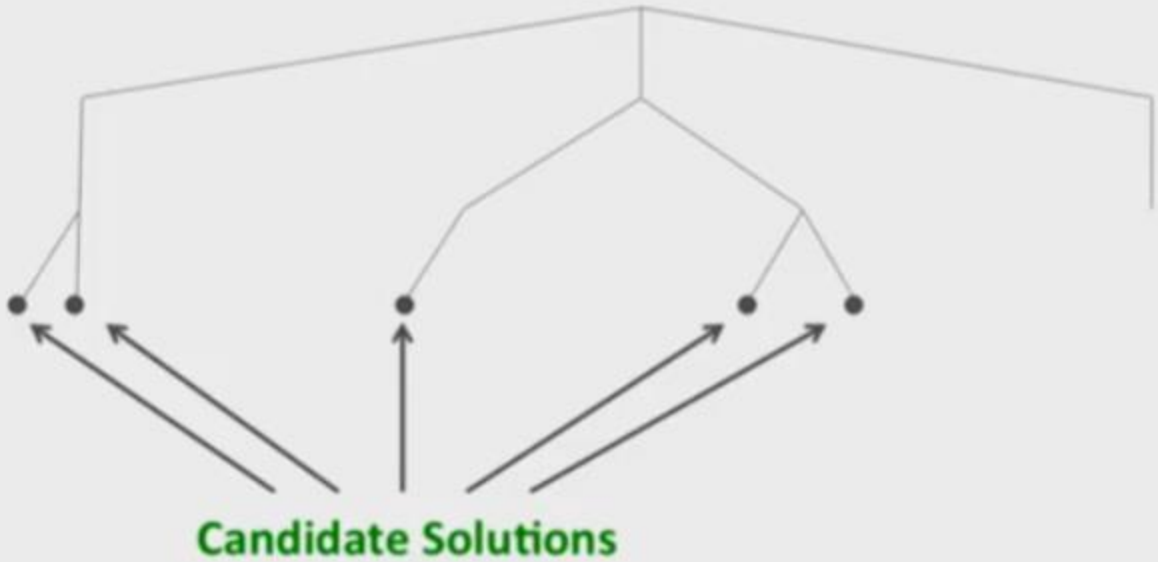
""	0
L	113
N	114
Q	128
E	129
LN	227
NQ	242
EL	242
QE	257
LNQ	355
ELN	356
QEL	370
NQE	371
NQEL	484

Branch-and-Bound Algorithms



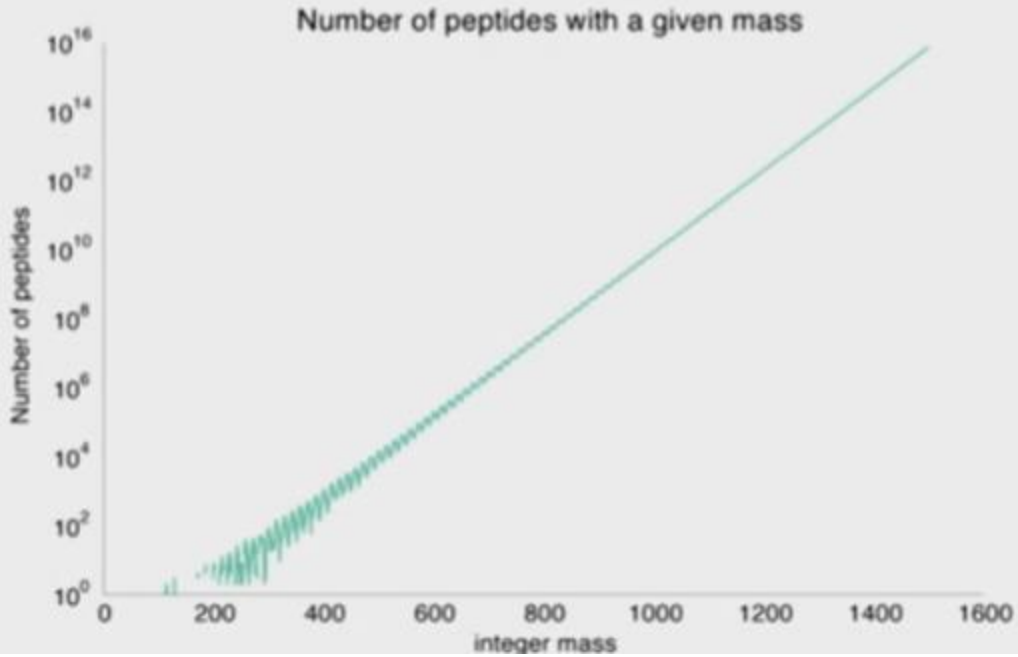


Branch-and-Bound Algorithms



Is This B-&-B Approach Efficient?

The brute force algorithm to cyclopeptide sequencing is **exponential**.



B-&-B for Cyclopeptide Sequencing

Spectrum	0	97	97	99	101	103	196	198	198	200	202
	295	297	299	299	301	394	396	398	400	400	497

G	A	S	P	V	T	C	I	L	N	D	K	Q	E	M	H	F	R	Y	W
57	71	87	97	99	101	103	113	113	114	115	128	128	129	131	137	147	156	163	186

FIGURE 4.6 Integer masses of amino acids. The integer mass of each amino acid is computed by adding the number of protons and neutrons in the amino acid molecule.

B-&-B for Cyclopeptide Sequencing

Spectrum	0	97	97	99	101	103	196	198	198	200	202
	295	297	299	299	301	394	396	398	400	400	497

Which amino acids have masses in *Spectrum*?

G	A	S	P	V	T	C	I/L	N	D	K/Q	E	M	H	F	R	Y	W
57	71	87	97	99	101	103	113	114	115	128	129	131	137	147	156	163	186

B-&-B for Cyclopeptide Sequencing

Spectrum

0	97	97	99	101	103	196	198	198	200	202
295	297	299	299	301	394	396	398	400	400	497

Extend these 1-mers into all possible 2-mers:

PA	VA	TA	CA
PC	VC	TC	CC
PD	VD	TD	CD
PE	VE	TE	CE
…	…	…	…
PW	VW	TW	CW
PY	VY	TY	CY

How can we **trim** this list?

B-&-B for Cyclopeptide Sequencing

Spectrum	0	97	97	99	101	103	196	198	198	200	202
	295	297	299	299	301	394	396	398	400	400	497

Which amino acids have masses in *Spectrum*?

G	A	S	P	V	T	C	I/L	N	D	K/Q	E	M	H	F	R	Y	W
57	71	87	97	99	101	103	113	114	115	128	129	131	137	147	156	163	186

G	A	S	P	V	T	C	I	L	N	D	K	Q	E	M	H	F	R	Y	W
57	71	87	97	99	101	103	113	113	114	115	128	128	129	131	137	147	156	163	186

FIGURE 4.6 Integer masses of amino acids. The integer mass of each amino acid is computed by adding the number of protons and neutrons in the amino acid molecule.

B-&-B for Cyclopeptide Sequencing

Spectrum

0	97	97	99	101	103	196	198	198	200	202
295	297	299	299	301	394	396	398	400	400	497

Extend these 1-mers into all possible 2-mers:

PA	VA	TA	CA
PC	VC	TC	CC
PD	VD	TD	CD
PE	VE	TE	CE
...	...	...	...
PW	VW	TW	CW
PY	VY	TY	CY

How can we **trim** this list?

B-&-B for Cyclopeptide Sequencing

Spectrum

0	97	97	99	101	**103**	196	198	198	200	202
295	297	299	299	301	394	396	398	400	400	497

PV is **consistent** with *Spectrum*:

Mass(P) = 97 Mass(V) = 99 Mass(PV) = 196

CD is **inconsistent** with *Spectrum*:

Mass(C) = **103** Mass(D) = **115** Mass(CD) = **218**

B-&-B for Cyclopeptide Sequencing

Spectrum

0	97	97	99	101	103	196	198	198	200	202
295	297	299	299	301	394	396	398	400	400	497

List of consistent 3-mers:

PVC	PVT	PTP	PTV	PCV
VPC	VPT	VTP	VCP	TPV
TPC	TVP	CPT	CPV	CVP

Expand, then **Trim**...

B-&-B for Cyclopeptide Sequencing

Spectrum

0	97	97	99	101	103	196	198	198	200	202
295	297	299	299	301	394	396	398	400	400	497

List of consistent 4-mers:

PVCP	PTPV	PTPC	PCVP	VPTP
VCPT	TPVC	TPCV	CPTP	CVPT

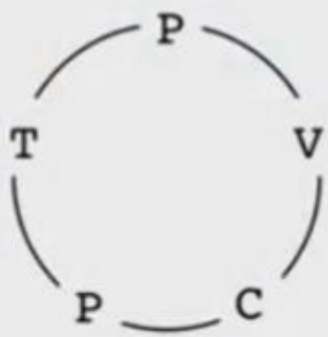
B-&-B for Cyclopeptide Sequencing

Spectrum

0	97	97	99	101	103	196	198	198	200	202
295	297	299	299	301	394	396	398	400	400	497

List of consistent 5-mers:

PVCPT	PTPVC	PTPCV	PCVPT	VPTPC
VCPTP	TPVCP	TPCVP	CPTPV	CVPTP



ONE CYCLIC PEPTIDE!

THEORITICAL SPECTRUM OF TYROCIDINE B1

0	97	99	113	114	128	128	147	147	163	186	227
241	242	244	260	261	262	283	291	333	340	357	388
389	390	390	405	430	430	447	485	487	503	504	518
543	544	552	575	577	584	631	632	650	651	671	672
690	691	738	745	747	770	778	779	804	818	819	835
837	875	892	892	917	932	932	933	934	965	982	989
1031	1039	1060	1061	1062	1078	1080	1081	1095	1136	1159	1175
1175	1194	1194	1208	1209	1223	1225	1322				

B-&-B for Cyclopeptide Sequencing

1. Find all amino acids whose masses occur in *Spectrum*. Add to *List*.
2. **Extend** each peptide in *List* by each of 18 different amino acid masses.
3. **Trim** inconsistent peptides from *List*.
4. Return any peptides in *List* whose theoretical spectra match *Spectrum*.
5. Iterate Steps 2-4 until *List* is empty.

From Theoretical to Noisy Spectra

Experimental spectra often produce errors.

Consider the following spectra for NQEL:

Theoretical:	0		113	114	128	**129**	227	**242**	**242**	257		355	356	370	371	484
Experimental:	0	**99**	113	114	128		227			257	**299**	355	356	370	371	484

False masses: present in experimental spectrum, absent from theoretical spectrum

Missing masses: present in theoretical spectrum, absent from experimental spectrum